Anthos Overview

NetApp Solutions

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Anthos Overview

Anthos with NetApp is a verified, best-practice hybrid cloud architecture for the deployment of an on-premises Google Kubernetes Engine (GKE) environment in a reliable and dependable manner. This NetApp Verified Architecture reference document serves as both a design guide and a deployment validation of the Anthos with NetApp solution deployed to bare metal and virtual environments. The architecture described in this document has been validated by subject matter experts at NetApp and Google Cloud to provide the advantages of running Anthos within your enterprise data-center environment.

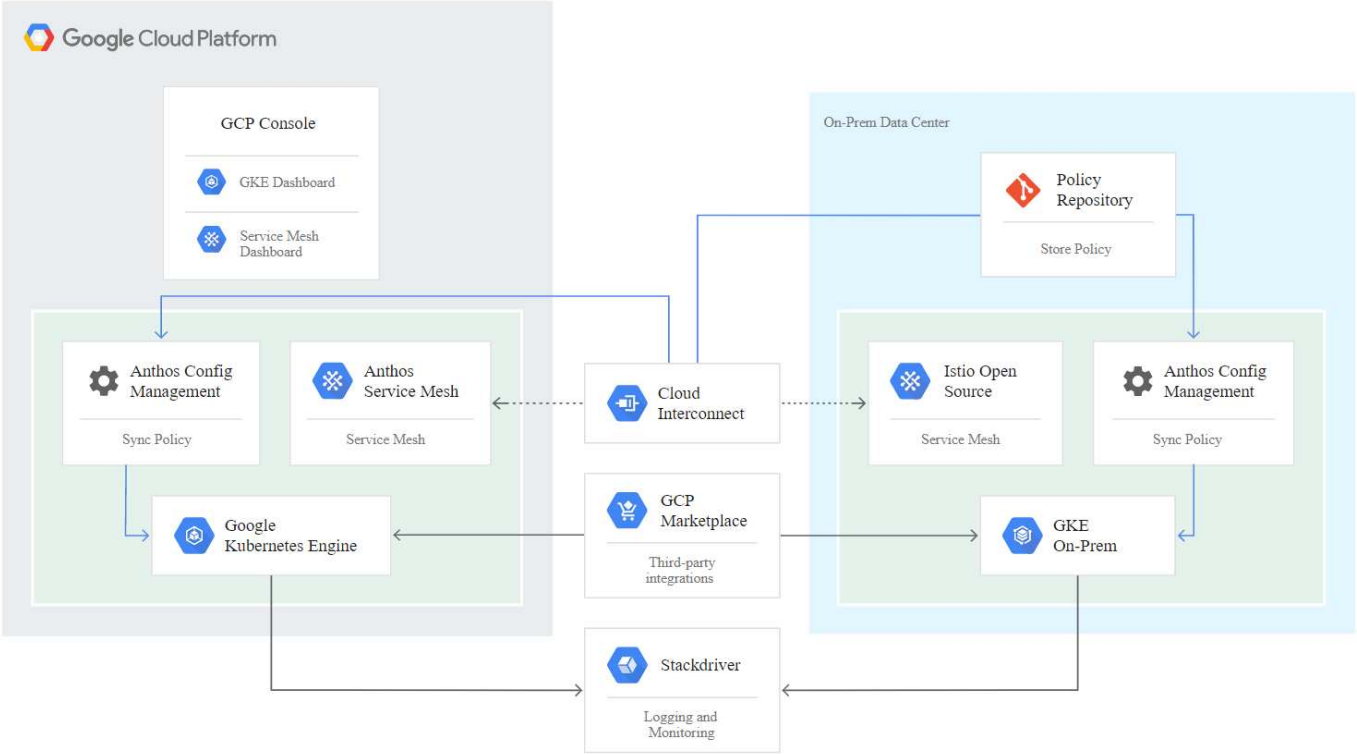
Anthos

Anthos is a hybrid-cloud Kubernetes data center solution that enables organizations to construct and manage modern hybrid-cloud infrastructures while adopting agile workflows focused on application development. Anthos on VMware, a solution built on open-source technologies, runs on-premises in a VMware vSphere-based infrastructure, which can connect and interoperate with Anthos GKE in Google Cloud.
Adopting containers, service mesh, and other transformational technologies enables organizations to experience consistent application development cycles and production-ready workloads in local and cloud-based environments. The following figure depicts the Anthos solution and how a deployment in an on-premises data center interconnects with infrastructure in the cloud.

For more information about Anthos, see the Anthos website located here.

Anthos provides the following features:

- **Anthos configuration management.** Automates the policy and security of hybrid Kubernetes deployments.
- **Anthos Service Mesh.** Enhances application observability, security, and control with an Istio-powered service mesh.
- **Google Cloud Marketplace for Kubernetes Applications.** A catalog of curated container applications available for easy deployment.
- **Migrate for Anthos.** Automatic migration of physical services and VMs from on-premises to the cloud.
- **Stackdriver.** Management service offered by Google for logging and monitoring cloud instances.



Deployment methods for Anthos

Anthos clusters on VMware

Anthos clusters deployed to VMware vSphere environments are easy to deploy, maintain, and scale rapidly for most end-user Kubernetes workloads.

For more information about Anthos clusters on VMware, deployed with NetApp, please visit the page here.

Anthos on bare metal

Anthos clusters deployed on bare metal servers are hardware agnostic and allow you to select a compute platform optimized for your personalized use case.

For more information about Anthos on bare metal clusters deployed with NetApp, visit here.

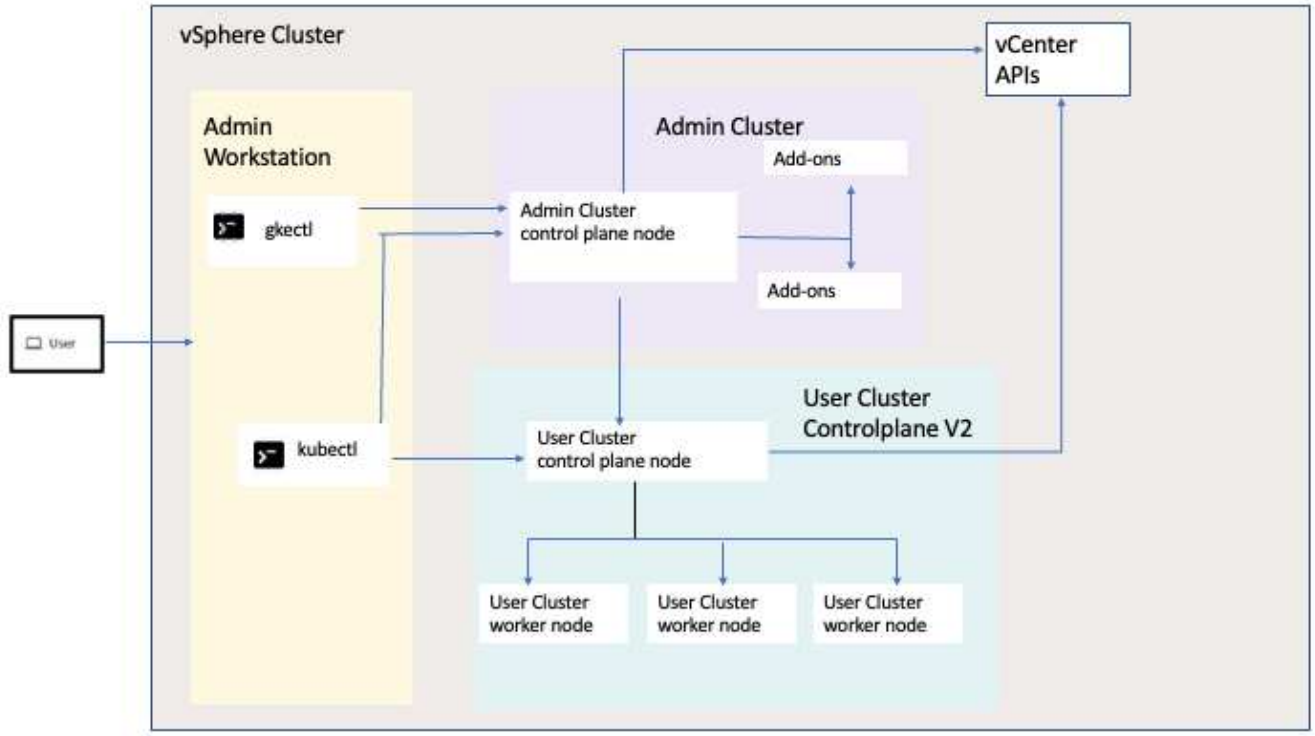
Anthos Clusters on VMware

Anthos clusters on VMware is an extension of Google Kubernetes Engine that is deployed in an end user's private data center. An organization can deploy the same applications designed to run in containers in Google Cloud in Kubernetes clusters on-premises.
Anthos clusters on VMware can be deployed into an existing VMware vSphere environment in your data center, which can save on capital expenses and enable more rapid deployment and scaling operations.

The deployment of Anthos clusters on VMware includes the following components:

- **Anthos admin workstation.** A deployment host from which `gkectl` and `kubectl` commands can be run to deploy and interact with Anthos deployments.
- **Admin cluster.** The initial cluster deployed when setting up Anthos clusters on VMware. This cluster manages all subordinate user cluster actions, including deployment, scaling, and upgrade.
- **User cluster.** Each user cluster is deployed with it's own load balancer instance or partition, allowing it to act as a standalone Kubernetes cluster for individual users or groups, helping to achieve full multitenancy.

The following graphic is a description of an Anthos-clusters-on-VMware deployment.



Benefits

Anthos clusters on VMware offers the following benefits:

- **Advanced multitenancy.** Each end user can be assigned their own user cluster, deployed with the virtual resources necessary for their own development environment.
- **Cost savings.** End users can realize significant cost savings by deploying multiple user clusters to the same physical environment and utilizing their own physical resources for their application deployments instead of provisioning resources in their Google Cloud environment or on large bare-metal clusters.
- **Develop then publish.** On-premises deployments can be used while applications are in development, which allows for testing of applications in the privacy of a local data center before being made publicly available in the cloud.
- **Security requirements.** Customers with increased security concerns or sensitive data sets that cannot be stored in the public cloud are able to run their applications from the security of their own data centers, thereby meeting organizational requirements.

VMware vSphere

VMware vSphere is a virtualization platform for centrally managing a large number of virtualized servers and

networks running on the ESXi hypervisor.

For more information about VMware vSphere, see the VMware vSphere website.

VMware vSphere provides the following features:

- **VMware vCenter Server.** VMware vCenter Server provides unified management of all hosts and VMs from a single console and aggregates performance monitoring of clusters, hosts, and VMs.
- **VMware vSphere vMotion.** VMware vCenter allows you to hot migrate VMs between nodes in the cluster upon request in a non-disruptive manner.
- **vSphere High Availability.** To avoid disruption in the event of host failures, VMware vSphere allows hosts to be clustered and configured for high availability. VMs that are disrupted by host failure are rebooted shortly on other hosts in the cluster, restoring services.
- **Distributed Resource Scheduler (DRS).** A VMware vSphere cluster can be configured to load balance the resource needs of the VMs it is hosting. VMs with resource contentions can be hot migrated to other nodes in the cluster to make sure that enough resources are available.

Hardware requirements

Compute

Google Cloud periodically requests updated validation of partner server platforms with new releases of Anthos through their Anthos Ready platform partner program. A listing of currently validated server platforms and the versions of Anthos supported can be found here.

Operating system

Anthos clusters on VMware can be deployed to both vSphere 6 and 7 environments as chosen by the customer to help match their current datacenter infrastructure.

The following table contains a list vSphere versions that have been used by NetApp and our partners to validate the solution.

Operating System	Release	Anthos Versions
VMware vSphere	8.01	1.15

Additional hardware

To complete the deployment of Anthos with NetApp as a fully validated solution, additional data center components for networking and storage have been tested by NetApp and our partner engineers.

The following table includes information about these additional infrastructure components.

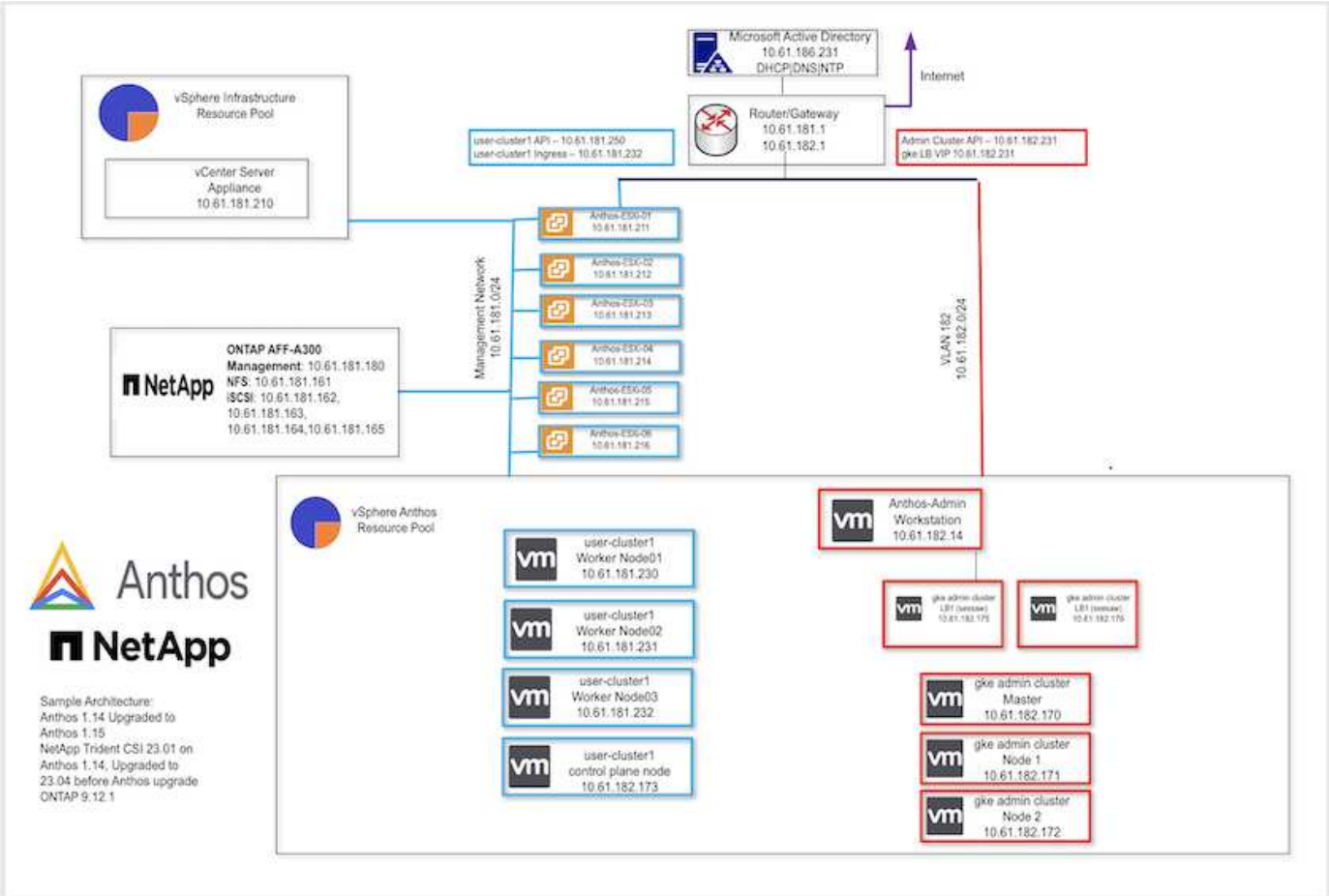
Manufacturer	Hardware Component
Mellanox	switch (data network)
Cisco	switch (management network)
NetApp	AFF Storage System

Additional software

The following table includes a list of software versions deployed in the validation environment.

Manufacturer	Software Name	Version
NetApp	ONTAP	9.12.1
NetApp	Astra Trident	23.04

During the Anthos Ready platform validation performed by NetApp, the lab environment was built based on the following diagram, which allowed us to test multiple scenarios using various NetApp ONTAP storage backends.



Network infrastructure support resources

The following infrastructure should be in place prior to the deployment of Anthos:

- At least one DNS server providing full host-name resolution that is accessible from the in-band management network and the VM network.
- At least one NTP server that is accessible from the in-band management network and the VM network.
- A DHCP server available to provide network address leases on demand should clusters need to scale dynamically.
- (Optional) Outbound internet connectivity for both the in-band management network and the VM network.

Best practices for production deployments

This section lists several best practices that an organization should take into consideration before deploying this solution into production.

Deploy Anthos to an ESXi cluster of at least three nodes

Although it is possible to install Anthos in a vSphere cluster of less than three nodes for demonstration or evaluation purposes, this is not recommended for production workloads. Although two nodes allow for basic HA and fault tolerance, an Anthos cluster configuration must be modified to disable default host affinity, and this deployment method is not supported by Google Cloud.

Configure virtual machine and host affinity

Distributing Anthos cluster nodes across multiple hypervisor nodes can be achieved by enabling VM and host affinity.

Affinity or anti-affinity is a way to define rules for a set of VMs and/or hosts that determine whether the VMs run together on the same host or hosts in the group or on different hosts. It is applied to VMs by creating affinity groups that consist of VMs and/or hosts with a set of identical parameters and conditions. Depending on whether the VMs in an affinity group run on the same host or hosts in the group or separately on different hosts, the parameters of the affinity group can define either positive affinity or negative affinity.

To configure affinity groups, see the appropriate link below for your version of VMWare vSphere.

vSphere 6.7 Documentation: Using DRS Affinity Rules.
vSphere 7.0 Documentation: Using DRS Affinity Rules.

> Anthos has a config option in each individual `cluster.yaml` file to automatically create node affinity rules that can be enabled or disabled based on the number of ESXi hosts in your environment.

Anthos on bare metal

Benefits

The hardware-agnostic capabilities of Anthos on bare metal allow you to select a compute platform optimized for your personalized use case and also provide many additional benefits.

Examples include the following:

- **Bring your own server.** You can use servers that match your existing infrastructure to reduce capital expenditure and management costs.
- **Bring your own Linux OS.** By choosing the Linux OS that you wish to deploy your Anthos-on-bare-metal environment to, you can ensure that the Anthos environment fits neatly into your existing infrastructure and management schemes.
- **Improved performance and lowered cost.** Without the requirement of a hypervisor, Anthos-on-bare-metal clusters call for direct access to server hardware resources, including performance-optimized hardware devices like GPUs.
- **Improved network performance and lowered latency.** Because the Anthos-on-bare-metal server nodes are directly connected to your network without a virtualized abstraction layer, they can be optimized for low latency and performance.

Hardware requirements

Compute

Google Cloud periodically requests updated validation of partner server platforms with new releases of Anthos through their Anthos Ready platform partner program. A listing of currently validated server platforms and the versions of Anthos supported can be found here.

The following table contains server platforms that have been tested by NetApp and NetApp partner engineers for the validation of Anthos on bare metal deployments.

Manufacturer	Make	Model
Cisco	UCS	B200 M5
HPE	Proliant	DL360

Operating System

Anthos-on-bare-metal nodes can be configured with several different Linux distributions as chosen by the customer to help match their current datacenter infrastructure.

The following table contains a list of Linux operating systems that have been used by NetApp and our partners to validate the solution.

Operating System	Release	Anthos Versions
CentOS	8.4.2105	1.14
Red Hat Enterprise Linux	8.4	1.14
Ubuntu	18.04.5 LTS (with kernel 5.4.0-81-generic)	1.14
Ubuntu	20.04.2 LTS	1.14

Additional hardware

To complete the deployment of Anthos on bare metal as a fully validated solution, additional data center components for networking and storage have been tested by NetApp and our partner engineers.

The following table includes information about these additional infrastructure components.

Manufacturer	Hardware Name	Model
Cisco	Nexus	C9336C-FX2
NetApp	AFF	A250, A220

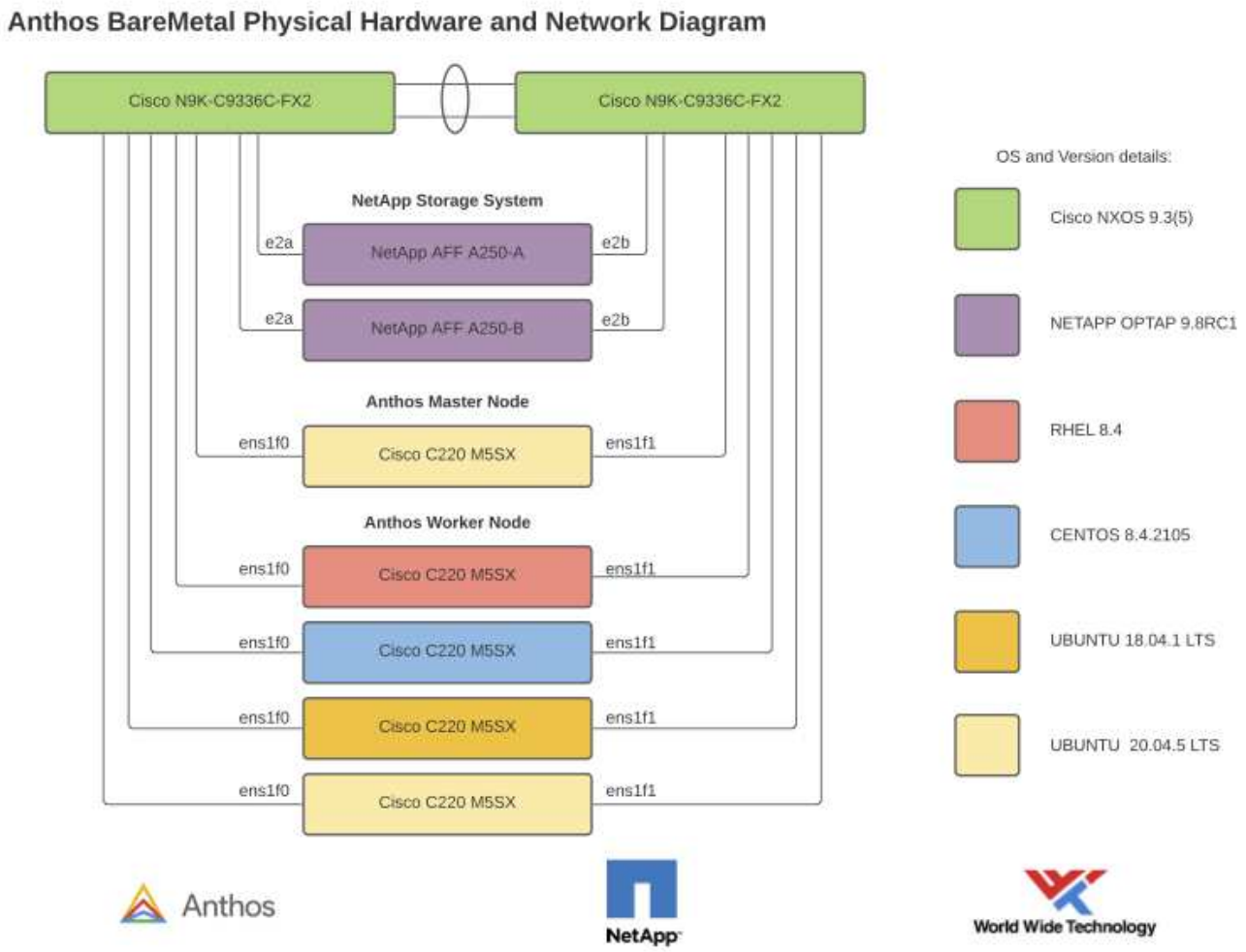
Additional software

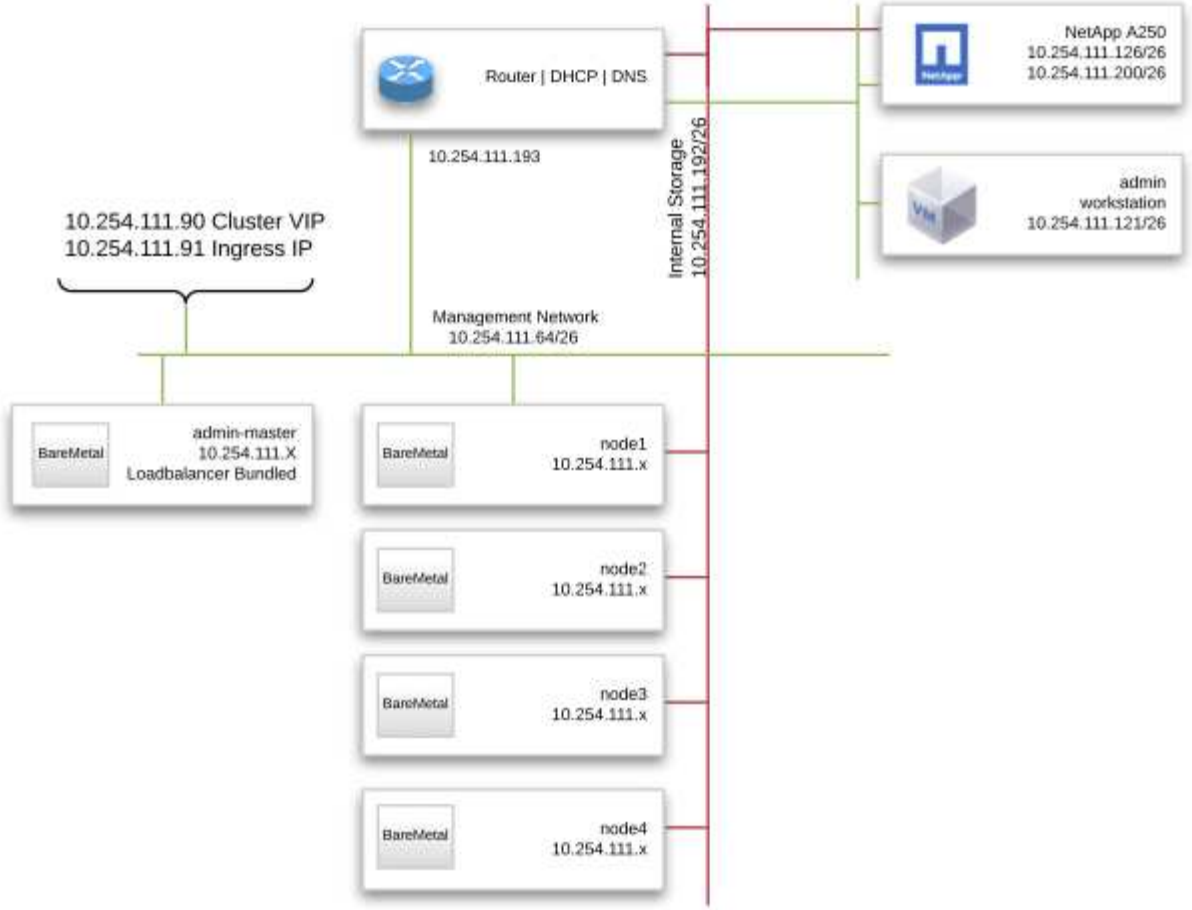
The following table includes a list of additional software versions deployed in the validation environment.

Manufacturer	Software name	Version
Cisco	NXOS	9.3(5)

Manufacturer	Software name	Version
NetApp	ONTAP	9.11.1P4
NetApp	Astra Trident	23.01.0

During the Anthos Ready platform validation performed by NetApp and our partner team at World Wide Technology (WWT), the lab environment was built based on the following diagram, which allowed us to test the functionality of each server type, operating system, the network devices, and storage systems deployed in the solution.





This multi-OS environment shows interoperability with supported OS versions for the Anthos-on-bare-metal solution. We anticipate that customers will standardize on one or a subset of operating systems for their deployment.

Infrastructure support resources

The following infrastructure should be in place prior to the deployment of Anthos on bare metal:

- At least one DNS server that provides a full host-name resolution accessible from the management network.
- At least one NTP server that is accessible from the management network.
- (Optional) Outbound internet connectivity for both the in-band management network.

There is a demo video of an Anthos on bare metal deployment in the Videos and Demos section of this document.

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